ECDLP lead discovery, 2026-10-24: prime-field index calculus first

Session report for RQ-ECDLP-361f3a (propose-ideas, lead-discovery round). Every statement below is labelled proposed, derived, measured (by whom) or independently verified. Nothing in this report was approved, dispatched or run as an experiment by this session (hard rule 3).

a. Orientation

- **Atlas brief (live tools, run from the CAR root at the commits below):** `ECDLP lead atlas (knowledge/leads/ecdlp/, reviewed 2026-10-24): 147 leads, 72 closed with a named obstruction, 51 open with a decided next step, 16 blocked, 8 unexamined. 2049/2078 ECDLP ledger records mapped to a lead. Changed since the review: crypto-autoresearcher: 146 paths; crypto: 16 paths; cryptanalysis: 32 paths; ecc-challenges: 3 paths.`
- **Delta status:** `--delta` lists 201 watched paths (54 decisions DEC-20261018..24, 19 CZLIFT/ICPERF/PEXT/SEMAEV hypotheses and evidence, 29 ideas IDEA-20261018..24, 12 findings/open problems, 19 problem records in CAR; 16 crypto paths; 32 cryptanalysis paths; 3 ecc-challenges paths). They were folded as **proposed row diffs** (section b and `FOLD.md`), not applied, and the review stamp was left alone (hard rule 3; AUD passes ask for diffs).
- **Commits worked at:** crypto-autoresearcher `ab0609ebee` (origin/main at 17:13 PDT; branch `lead-discovery/20261024-mythos-ecdlp`); crypto `5cb553810e` (origin/main, detached worktree); cryptanalysis `c5aa7b39b` (origin/main, detached worktree); ecc-challenges `df8cd52` at the first fetch, `86f8a57` by the end of the session; crypto-lean: **not checked out** on this host (no Lean imports were taken from it).
- **Research question:** RQ-ECDLP-361f3a (new; minted with `tools/allocate_id.py`, `--check` OK).
- **Tools:** live (`lead_atlas.py --brief/--delta/--coverage/--check --strict-paths/--render/--match`, `build_frontier_map.py --out/--match/--check`, `obstruction_registry.py --unexamined/--debt`, `build_ledger_index.py`, `validate_ledger.py`, `allocate_id.py`), not a snapshot.
- **Baseline read (crypto, origin/main, read by a delegated reader and spot-checked):** Table B prime ladder fp10..fp24 at S_IC/S_rho 5.47x (log2 r 11.9) to 13.5x (23.4), validated 13/13, no sealed bound on any Table B row; sealed prime toy IC bounds `ECBND1h93b7a9d5c187` (ic.pipeline mitm:negation_folded, alpha 0.8407 [0.6459, 0.9528]) and `ECBND1h9879c0aa348f` (subtract, alpha 1.0054 [0.8805, 1.0769]) against `rho.negation` `ECBND1h9896c8176a62` (alpha 0.4545 [0.3876, 0.5927]); "Prime curves have no algebraic oracle in ecbench"; Koblitz Table A at rule v8: rho 4.67-18.07x faster cold at n = 41..61, no sealed Koblitz IC bound (all six inadmissible); no end-to-end IC row on the m = 83 gate curve.

b. Audit table

Full per-lead record: `AUDIT.md` (seven family audits, every finding quoting the row's obstruction.statement or next_action and an evidence path). Fold of the 201 changed paths: `FOLD.md`. Counts (findings that propose a change; "no change" rows stay in the files):

family	leads	findings	Lean target statements (AUD-3/4)
generic	26	25	7
ic-binary	21	21	8
ic-engine	19	23	4
ic-prime	24	28	7
transfer	21	30	4
structure	17	17	9
exotic	19	16	7
total	**147**	**160** (AUD-1 21, AUD-2 52, AUD-3 21, AUD-4 55, AUD-5 9, AUD-6 2)	**46**

Fold (section a): 197 of 201 paths read; 182 advance an existing lead (row edits proposed in `FOLD.md`), 15 are not about ECDLP (coverage `NONE`), 4 unclear; **6 new mechanisms** no lead covers, proposed as new rows for the Coordinator (not filed here because each needs its own `--match` and criteria block): `LEAD-ICP-fp4-gaudry-d4-ladder` (IDEA-20261018-2c37f5), `LEAD-TRF-cubic-ghs-genus3-hyperelliptic-2lp` (IDEA-20261018-670088), `LEAD-ICP-pkm-tower-algebraic-oracle` (14 `research/pkm_*` directories cited by no lead), `LEAD-TRF-prime-degree-ghs-genus-floor` (GOAL-PEXT-4a1d6b), `LEAD-EXO-koopman-transfer-operator` (KN-OPEN-010), and `LEAD-TRF-torsion-section-digit-fourier` (KN-OPEN-d3a6f0, split from LEAD-TRF-padic-canonical-section). Two fold rows propose disposition changes on LEAD-TRF-padic-canonical-section (DEC-20261023-2e7447 conditional reopen; DEC-20261023-36b4f0 confirmation).

Disposition changes proposed (21) and their adversarial verification (3 skeptics each, majority rule):

lead	pass	proposed	verification
LEAD-RHO-non-automorphism-equivalence	AUD-5	the queued proof target is too weak to close the lead; sharpen the first step	**survives** (3 votes)
LEAD-ICE-chained-semaev-dreg-vs-semiregular	AUD-4	KR-IC-fbdb61 already says semi-regular predictions do not describe descended systems, so only the block/topology-matched null is a valid co-arm; rewrite the step	**survives** (3 votes)
LEAD-RHO-cheon-deployed-applicability	AUD-4	a bound without factorisation decides the discriminator	refuted
LEAD-ICB-factor-base-subspace-choice	AUD-1	revisit trigger fired for the product-profile property	refuted
LEAD-ICB-tau-horner-chain	AUD-4	a counting identity decides the m! accounting before the run	refuted
LEAD-ICE-sat-xor-pdp-oracle	AUD-4	recorded counts already exclude “priced S/rho below 1 across the band”	refuted
LEAD-ICE-first-fall-degree-assumptions	AUD-4	the counterexample branch is realised on a finite model (Lean certificates)	refuted
LEAD-ICP-lattice-ic-relation-construction	AUD-4	two of three routes reduce to standing obstructions	refuted (the evidence path is a proposal report, not a deciding record)
LEAD-TRF-padic-canonical-section	AUD-1	the clause “any non-homomorphic section recovers d only at the random 1/p	**survives** (3 votes; all three skeptics give the same split, see PROPOSED-DIFFS.md 7)
[illegible] transfer	[illegible]	open_next[illegible] By DFC-20261024-23d64d; split the row, keep the homomorphic/ formal-log lit: clause closed (kind theorem	[illegible] DFC-20261024-23d64d

Net: **8 of 21 proposed disposition changes survive** three skeptics each; 13 are refuted, in every case because the cited record is a plan, a proposal, a classification or an un-archived decision rather than a deciding record. The 8 survivors are written as proposed diffs with the skeptics' corrected wording (`PROPOSED-DIFFS.md`, items 7-12); none is applied.

Prime-field findings that change what to do next (ic-prime, ic-engine; the proposer's wording, with the evidence path in `AUDIT.md`):

- AUD-3, `LEAD-ICP-extension-field-embedding`: a structural lemma replaces the cost model: for odd n, `{P in E(F_{p^n}) : x(P) in F_p}` **is** `E(F_p)` (y^2 in F_p forces y in F_p), and for even n it is `E(F_p)` plus the twist; the subfield base is the original group, so the embedding buys nothing by construction. This is stronger than the exponent inequality proved in `ExtensionFieldEmbedding.lean` and is the better Lean target (statement in `AUDIT.md`; not written this session).
- AUD-3, `LEAD-ICP-generic-table-decomposition` and `LEAD-ICE-generic-oracle-arity-cost-floor`: the m <= 3 part of the "theorem" is a two-term cost model; the audit states the exact (constant-level) AM-GM form `a/B^(m-1) + B^2 >= ((m+1)/(m-1))((m-1)a/2)^(2/(m+1))` with `a = m! N T`, sub-rho only if `T = O(N^((m-3)/4))`. The exponent form is proved in `ArityOracleThreshold.lean`; the AM-GM form with constants is the natural follow-up target.
- AUD-1, `LEAD-ICE-generic-oracle-arity-cost-floor`: revisit clause (b), "a non-generic yield exceeds C(B+m-1,m)/N on average", **cannot fire** because KN-FIND-007 proves the mean is exactly that for every base of size B; rewrite it as a target-sampling rule with O(1) GAE per sampled target and a conditional yield above the mean (this is the H1 escape in the threshold theorem's own terms).
- AUD-2, `LEAD-ICP-generic-table-decomposition`: three forecloses phrases cover coordinate-reading oracles the generic bound does not reach; qualify them ("with a generic oracle, or at m <= 3 with any oracle under the AUD-3 lemma").
- AUD-2, `LEAD-ICP-representation-subset-sum`: scope "any representation/filter variant" contradicts the row's own revisit_when; restrict to filters that use only group operations and equality of encodings.
- AUD-2, `LEAD-ICP-algebraic-membership-factor-base`: the prime-field theorem is cited by unqualified phrases ("multiplicative subgroup factor base x^B = 1") that block the binary variant IDEA-20261018-329d0a; qualify "over F_p".
- AUD-2, family misfilings: `LEAD-ICP-icperf-writability-boundary` (binary scope, filed in ic-prime; and its chained-Semaev phrase blocks the very representation its revisit_when names), `LEAD-ICP-icperf-binary-pdp-solver` (overlaps LEAD-ICE-sat-xor-pdp-oracle), `LEAD-ICP-retained-setup-online` (Koblitz n61/n83 rows duplicating LEAD-ICB-retained-setup-online; no prime-field row exists).
- AUD-4, `LEAD-ICP-residual-partial-relations`: the exponent is already derived (H-PFDR-cb9606): work ~ N^((1+beta)/2), LA balance at beta = 1/3 gives 2/3 against the fitted 0.841 [0.646, 0.953]; add "fit against 2/3" to discriminates (this is the k = 1/2 point of the threshold theorem: an oracle of order 1/2 at m = 2 gives 2/(m+1-k) = 2/2.5 = 0.8, so the derived 2/3 needs the large-prime accounting the model does not have; both numbers stay above rho).

- AUD-4, `LEAD-ICP-prime-ic-constant-engineering`: the validation-gate half of the next step is done (LEADERBOARD lines 49-56 mark Table B validated; DEC-20261023-122e32 accepted EXP-ICPERF-6c546e); compute the counted batching gain from calibration.json and run the paired arm against both the incumbent recipe and a batched-rho reference.
- AUD-4, `LEAD-ICP-coincidence-harvest-structured-base`: EXP-PFDR-bb0d26 has run (DEC-20261022-9aec2b: O-NULL on 44 of 45 powered cells, max |kappa-1| = 0.0414, planted kappa >= 1.2885 detected); drop the "run it" clause, add its independent review and the a0830c rungs at 32/36 bits.
- AUD-4, `LEAD-ICP-coppersmith-smallroot-semaev`: no strategy-independent reach bound is known, so the queued shift-polytope computation is the decider; absorb IDEA-20261015-9defe9 (Newton-target shifts) into it. The lattice scout (section c) adds the staleness flag that the row quotes numbers superseded by CORR-20261024-b2a799 (eps(S_3) = 0.2803, margin −0.6086 at (4,2)).
- AUD-4, `LEAD-ICE-charged-toy-ic-exponent-vs-rho`: measure the O5 oracle's per-query cost slope before building the pipeline; slope >= (m-3)/4 at every arity closes the lane on exponents (the same conclusion as PROPOSED-DIFFS.md section 2, reached independently by the audit agent from KN-FIND-007).
- AUD-4, `LEAD-ICE-gpu-rootfind-oracle`: an Amdahl floor from the validated Table B phase shares puts the matched GPU ratio at >= 1 at fp12, fp16, fp18, fp20 even with a free relation phase; restrict the session to fp10, fp14, f2m18, f2m21 and map kernel efficiency below h_min to closed_obstructed (cost_model).
- AUD-4, `LEAD-ICE-relation-graph-rank-large-prime` and `LEAD-ICE-semaev-yield-model-coset-typing`: KN-FIND-007 forces the gain to come from the oracle side (LA is < 0.5% of Table B S); the typed-base gain is (m-1)!, not m!, at matched per-variable dimension and loses m^m at matched total base.
- AUD-4, `LEAD-ICE-hybrid-guess-determine-l7`: both E8 outcomes stay above rho in column touches (+8.43 bits constant k; +30.7 to +58.3 growing k); gate E8 on the unmeasured touch-to-GAE conversion K.
- AUD-2, scope qualifiers needed on `LEAD-ICE-semaev-cover-monodromy` (F_p, p > 3 only), `LEAD-ICE-boolean-f4-f5-f6-engines` and `LEAD-ICE-semaev-chain-cost-model` (binary, Semaev 2015 cost model), and `LEAD-ICE-descent-closure-degree-at-scale` (random subspaces only; the polynomial-basis V of KR-IC-73db3f was never run).

Binary / Koblitz (ic-binary, 21 findings; `AUDIT.md`): the family audit found eight short-proof candidates (the conservation lemma LT-1 used by three rows, the tau-Horner bijection LT-7, the cofactor-filter lemma LT-6, the writability/alpha inequality LT-2 "there exists l with n + l(1 + alpha - m) < n/2 iff alpha < m/2 - 1", the affine-invariance of degree-d spans LT-3, the scalar-fixed-point lemma LT-4, the LA-share bound LT-5); its only proposed disposition changes (factor-base-subspace trigger, tau-Horner counting) were refuted by the skeptics, so the binary rows stand as they are.

Cross-session input, not verified here: a parallel desk review on this host (memory note `ecdlp-factor-base-arity-review-2026-10`, 2026-10-24) reports four gaps the repos do not record: no algebraic solving degree exists at m >= 5 in any repo (first chained cells (5,2,11),

(5,3,13), (6,2,11) fit the 64-variable engine); the literature's degree-4 flat was measured at large overdetermination Delta = n - m*l while the repo's rise at l = 7 is at Delta about 0 (a Delta-sweep at fixed (m, l) is its top-ranked experiment); Galbraith-Gebregiyorgis 2014 Experiment 2 (m = 4, l = 4 binary Edwards invariants) was never built; the symmetric e_k presentation above the l = 6 crossover was never run. These are recalled from the note, not from a committed record, and belong to the ic-binary and ic-engine next_actions once checked.

c. Graveyard

Full record: `GRAVEYARD.md`; dossiers and raw gate output: `scouting/`. Twelve outside fields were scouted (coding theory and ISD, lattice small-root methods, SAT/XOR and planted systems, additive combinatorics, commutative algebra, p-adic and tropical geometry, arithmetic statistics, query and communication complexity, HPC constants, algebraic geometry of covers, the polynomial method, algorithmic number theory). Screened count: **47 candidates proposed after 152 scout-level kills** (every kill with its reason in `GRAVEYARD.md` section D). Gate outcome for the 36 candidates that received a verdict: **8 killed (7 N1 duplicates of a lead, 1 N2 known result), 28 judged `advances_lead` (a sharper step on an existing open lead), 0 new mechanisms.** Eleven candidates have no gate verdict (the gate agents hit the session limit twice) and are listed as unexamined. No skeptic votes were obtained (every refutation agent failed on the limit), so nothing below is independently refuted or confirmed; the lead author's own reading is marked as such.

candidate	killed by	obstruction kind	forecloses phrases	record needed
Riemann-Roch excess lemma: no AG-code decoder or ISD algorithm is a point-decomposition oracle	N1 (duplicate of LEAD-EXO-ideas-corpus-algebraic-transforms; FINDING-PF-IC-001 P1539, rejected ideas 013/014/130/…)	proof (the lead's)	AG code decoding as decomposition; list decoding the elliptic code; syndrome decoding for ECDLP	none new; the lead's record
Function-field corner lemma: F_2[z] Popov-form lattice decomposition on ECC2K-130	N1 (same mechanism as LEAD-ICP-copper-smith-small-root-semaev, binary arm)	cost_model	polynomial-lattice small roots of S_m over F_2[z]; Popov form decomposition; function-field Coppersmith	that lead's reach record
Exact tree-like Res(xor) optimum of UNSAT descended PDP instances	N1 (LEAD-ICB-decomposition-encoding-and-sat, closed; candidate does not escape its measured alpha ~ m)	measured	parity decision tree for PDP; tree-like Res(xor) refutation size; UNSAT core size of descended S_m	the lead's EV records
Popcount-filtered exact refutation-degree meter for squaring and Chebyshev towers	N1 (LEAD-ICP-algebraic-membership-factor-base; residual item is IDEA-20261018-e0857e)	theorem	refutation degree of tower bases; Chebyshev tower factor base; squaring tower membership	KN-FIND-007 / the lead's record
Valuation-triviality lemma: p-adic tropical initial-form decoder for a Semaev system	N2 (KR-IC-5931ee consequence; frontier delta 0)	literature	p-adic valuation decoder for Semaev; tropical initial forms of S_m; Newton polygon decomposition	KR-IC-5931ee
(tau, mu) admission region for IC decomposition oracles, find-one vs list-all	N1 (quantitative complement of LEAD-ICE-generic-oracle-arity-cost-floor; now covered by next action;	proof (model)	oracle admission region; find-one vs list-all; tau-mu exponent plane;	TASK-20261024-9fbb0f
SAT/XOR oracle admission region: bearing for IC decomposition oracles, find-one vs list-all	N1 (the deprecated HBC-IDEA-20261024-f451c49 step	n/a (open lead)	boundary bounding the SAT or a non-isomorphism oracle; tau-mu exponent plane;	IDEA-20261024-f451c49
Fourier-coefficient transfer, the Miller cocycle	N1 (duplicate of LEAD-ICE-non-Kummer-transfer-already-holds or absolutely-irreducible-now-covered-by-IDEA-20261024-f451c49)	7	Galois-group addition; the Miller	

Scout-level kills worth recording because they close routes a future session would otherwise retry (all in `GRAVEYARD.md` D, with the scout's argument): every list-decoding, ISD, folded-code, interleaved and quasi-cyclic decoding route (agreement equals the dimension; collisions reduce to the group-law MITM; no small-order translation on a prime-order group); every Coppersmith variant beyond the corner lemma (Jochemsz-May, Herrmann-May, Bauer-Joux, Kannan embedding, BKZ at any dimension, rank-r GAP lattices, Farey bases, Bombieri-Pila); sum-product, incidence, Kloosterman/Weil/Burgess and Balog-Szemeredi-Gowers routes (vacuous below sqrt(p) or above the break-even); overlap-gap, low-degree, k-XORSAT threshold, cutting-planes and lifted Res(xor) bounds (wrong model or gadget-composed formulas); Mordell-Weil CVP, Weil polynomial enumeration for Jacobian carriers and EDS ratio stripping; and every binary/Koblitz coding-theory route (reduce to SO-QSP, rank-metric K7 or SO-FROB).

d. Survivors, ranked

No candidate opens a new lead; the six carried forward are sharper first steps for existing open leads, filed as IDEA records with `prior_art.lead` set, criteria blocks and draft protocols in their notes (section g). Ranking by P(first step rules it in) x gain / cost, with each factor the lead author's estimate (unverified by a skeptic; the dossier's own self-kill risk is quoted):

idea (advances lead)	tier	claim type	P(rule-in)	gain if it does	cost of the step	first step
IDEA-20261024-741c49 conflict-priced 5a band for the native-XOR SAT oracle from the recorded ecbench counters (LEAD-ICE-sat-xor-pdp-oracle; ECC2K-130 family, n = 17, m = 2)	I	exact_computation + measured	the arithmetic stands (every number traced to `research/ecbench_sat_pdp_20261009a`); the ORDER is **unresolved** under crypto/AGENTS.md 5a: SAT/strong-rho is 0.854 [0.825, 0.883] at w = 0 but 7.34 at w = 0.1 and 51.0 at w = 1, break-even w* = 1/458 = 2^-8.84, so the lead's sub-rho branch holds only if a CDCL+XOR conflict costs under 1/458 of a group addition; 0.3 that a qualified-host calibration lands (be-	decides the one binary row with a sub-1 unpriced ratio; re-planned ns-per-conflict calibration only if the calibration itself is the deciding step	zero for step (i); step (ii) needs a re-freeze (PROTOCOL-4: n = 17 and 23, l-ladder 8-12, conflict budget 300,000, since n = 19 and 29 degenerate and n = 23 trips the frozen 20,000 budget)	file the band computation as the repricing; an admitted ns-per-conflict calibration on a qualified host; the PROTOCOL-4 re-freeze
IDEA-20261024-[illegible] exact-[illegible]	I (rule-out)	exact_computation + measured	the [illegible] (good-	[illegible]	1-2 days	[illegible]

Four more gate-passed candidates are proposed as next_action sharpenings without an IDEA record (section b's proposed diffs or `GRAVEYARD.md` B): the exact all-target decomposition census from the log-domain spectrum (LEAD-ICE-decomposition-probability-audit), the cofactor-coset counting law that makes the volcano l = 2 yield effect 2-torsion bookkeeping (LEAD-STR-class-representative-ic-selection), the Table B "rho denominator" decomposition with a falsifiable 28/32-bit projection (LEAD-RHO-matched-rho-calibration), and the Kummer-conductor / Weil-Deligne bound for the Miller primitive (LEAD-TRF-miller-cocycle-nonhom-transfer).

Two corrections the record writers made to the lead author's first reading, kept here because they change how the ideas are read: (1) the full-sumset arm (IDEA-20261024-06f16f) is a **k = 0 oracle** (one lookup per walked target), not a k = ceil(m/2) one; its exponent stays at m/(2m-1) > 1/2 only because the B^m table build is charged, which is the arity-2 memory-coupling theorem of `MemoryCouplingArityTwo.lean` generalised to arity m (proposed as that idea's Lean target); and (2) the occupancy calculator (IDEA-20261024-892d61) settles only the two occupancy contrasts of EXP-ECTD-0cf008, not all 17 missing arms; its rank-prefix, interval, replay and transported-base-equality exit criteria stay owed. The writers also found two crypto study directories cited by no lead (`research/ic_triple_table_20260923`, `research/ic_triple_counted_20260923`), an atlas gap for the next fold.

What the round says about the prime-field path forward. Every candidate that survived the gates is either a closure (corner lemma, IPS degree floor, writability floor, occupancy calculator) or a constant-level engineering step (the tau = 0 sumset arm). Together with the threshold theorem (section f) the prime-field picture is: the exponent question is open only for a decomposition oracle of enumeration order k < (m-1)/2, every lattice, Groebner-on-digits, SAT and enumeration route proposed in this round sits at or above the line by its own dossier, and the measurable work is (1) the oracle-order census, (2) the tau = 0 sumset arm against Table B, and (3) sealing the three closures as Lean targets.

e. Obstruction matrix and raw `--match` output

For IDEA-20261024-a2b05d (which advances a closed lead rather than opening one), the raw gate output is in `gates/gates-arity-threshold.txt`; the top hits were `LEAD-ICE-generic-oracle-arity-cost-floor` (24, 14, 24 on the three phrasings), `LEAD-ICP-generic-table-decomposition` (20), `LEAD-ICE-rootfind-pairtable-oracle` (20), `LEAD-ICP-coppersmith-smallroot-semaev` (17); frontier hits `KR-IC-5931ee` (19), `KR-RHO-13bf67` (15); ledger hits IDEA-20260803-fa9839, IDEA-20260808-2ef5c8, IDEA-20260829-b5a2c8, IDEA-20260830-3c0861, IDEA-20260830-4ab603 (the two-term model lineage). Verdict: same mechanism as the first hit; the idea upgrades its closure (AUD-3) and does not file a new lead.

f. Lean status

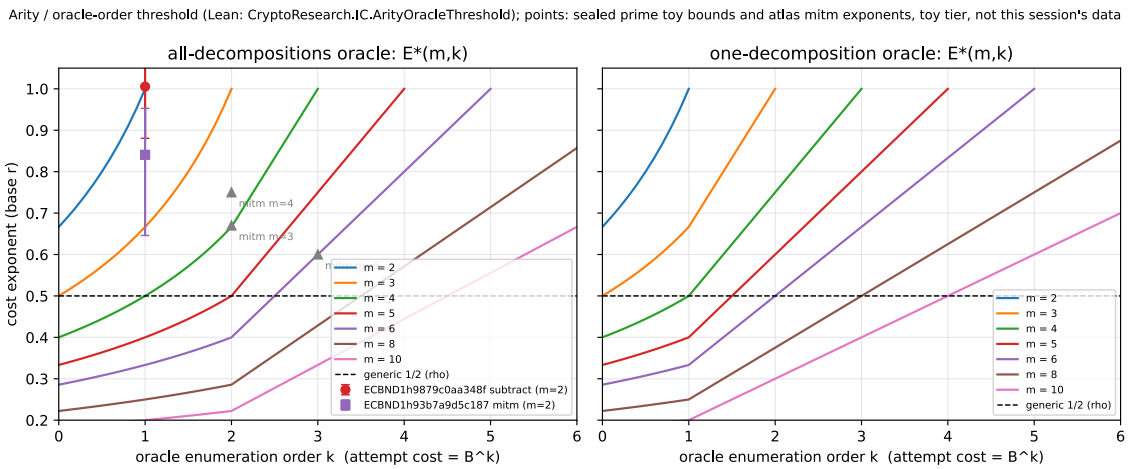


Figure 1: Arity / oracle-order threshold: cost exponent E*(m, k) against the generic 1/2 for both oracle semantics, with the sealed prime toy IC bounds and the atlas mitm exponents placed at their model order (editable source: render.py; nothing in the figure is a measurement produced by this session).

theorem (file)	file or attempt dir	role	status	open bridges
`CryptoResearch.IC.AritySeparation` (19 theorems: `costAll_ge_small`, `costAll_ge_large`, `costAll_ge_one`, `small_threshold_le_large_threshold`, `free_oracle`, `linear_oracle`, `quadratic_oracle`, `mitm_never_sub_generic`, `exhaustive_ge_one`, `attemptsAll_le_attemptsOne`, `costAll_le_costOne`, `costOne_ge_small`, `costOne_small_attained`, `costOne_ge_enum`, `costOne_enum_attained`, `main_hyps_satisfiable`; controls `needs_H1..H4`)	`formal/CryptoResearch/IC/AritySeparation.lean`; target `formal/targets/ic-arity-oracle-threshold-20261024.yaml` (TASK-20261024-9fbb0f)	cost-model algebra (rule-out within the model H1-H4)	compiles under plain `lean` v4.35.0-rc1 / Mathlib 7fa22735: 0 errors, 0 `sorry`, every `#print axioms` = propext, Classical.choice, Quot.sound; **repository gate not run to completion** (problems/formal-lean-external-volume-eintr.md); `needs_semantic_review = true`	H1-H4 are hypotheses; no bridge to curve points, counts or finite sizes; the one-decomposition semantics for 1 < k is not interpolated (only k <= 1 and k >= 1 bounds)
`CryptoResearch.IC.MemoryCouplingArityTwo` (7 theorems; control)	`formal/CryptoResearch/IC/MemoryCouplingArityTwo.lean`; target `formal/targets/ic-memory-coupling-arity-two-20261024.yaml` (TASK-20261024-2be6ac)	cost-model algebra (WOLE-20260830-4ab603 correction)	same: compiles under plain `lean`, 0 errors, 0 `sorry`, standard axioms; gate not run	H5 (B^2 table) is a hypothesis; unstructured bases only

`CryptoResearch.IC.ExtensionFieldEmbedding` (AUD-3 upgrade for `LEAD-ICP-extension-field-embedding`, kind cost_model) | `formal/CryptoResearch/IC/ExtensionFieldEmbedding.lean`; target `formal/targets/ic-extension-field-embedding-20261024.yaml` (TASK-20261024-551bd9) | cost-model algebra | compiles under plain `lean` (see `checks/`); gate not run | H6 (Gaudry's heuristic cost p^(2-2/n+o(1)) on E(F_{p^n}), KR-IC-46d822) is a hypothesis |

Derivation for the third target (source of its claim). Embedding the prime-field problem on E(F_p) (order about p) into E(F_{p^n}) and running Gaudry's decomposition with the subfield base {x in F_p} costs, under H6, p^(2-2/n+o(1)) for every fixed n >= 2 (KR-IC-46d822: q^(2-2/n) with q = p). The reference is rho on the original group at p^(1/2). Claim: for every real n >= 2, 1/2 < 2 - 2/n; equivalently there is no n >= 2 with 2 - 2/n <= 1/2, so the embedding enlarges the problem faster

than the subfield base helps, at every extension degree. Control: without the hypothesis n >= 2 the statement fails at n = 1 (2 - 2 = 0), which is the trivial extension and not an embedding, so the degree hypothesis is load-bearing. Non-vacuity: n = 2 gives the exponent 1, which is the LEAD row's own "p at n = 2". Proved in the file as `gaudry_exponent_gt_half`, `embedding_never_helps`, `needs_degree_two`, `main_hyps_satisfiable`.

Fidelity items (sheriff B3 a-e), all files: (a) statement paired with the informal claim clause by clause in the file header and in `IDEA-20261024-a2b05d.notes.md` sections 3 and 8, forall/exists order written out in the IDEA's `quantifier_order`; (b) `main_hyps_satisfiable` in each file; (c) no new computable definitions (the definitions are real-valued `max` expressions; no `#eval` differential test applies); (d) no data enters; (e) scope: real m >= 2, k >= 0, beta > 0, exponents only; never cited at any finite size.

g. Experiment specs awaiting approval (approved_by: null)

1. `specs/EXP-SPEC-oracle-order-census.md` - prime-field decomposition-oracle order census (fitted k per implemented oracle against the threshold line; positive controls with k = 1 by construction; matched rho.negation; A/A).
2. Draft PROTOCOL for each carried-forward idea's first step, in section 4 of its notes file (`ledger/proposals/<IDEA-ID>.notes.md`), with the criteria block in section 3: IDEA-20261024-1b76a4 (corner-lemma review, rank-2 GAP probe, cuso on S_3), IDEA-20261024-e65d46 (five-lemma proof review, exact polynomial-calculus degree checker at s = 6..14, three toy primes), IDEA-20261024-5bbcda (sampled-coefficient certification of H4w at 128/256 bits), IDEA-20261024-741c49 (band computation from the recorded counters; PROTOCOL-3 re-run with the 0 / 0.1 / 1 GAE band), IDEA-20261024-06f16f (paired `mitm-full:m=3` arm against the Table B incumbent at fp20/22/24/28 with matched rho.negation and A/A), IDEA-20261024-892d61 (pre-registered exact occupancy tables for the EXP-ECTD-0cf008 recovery amendment). All carry `approved_by: null`; none was compiled, relayed or run.

h. Blockers (problems/ records)

- `problems/formal-lean-external-volume-eintr.md` (new): Lake/`lean` reads on the external volume fail intermittently with EINTR; the Lean gate could not be run to completion; the theorems compile under plain `lean` with retries.
- `problems/README.md` is held by Conductor task T-5065 (another session), so the index line for the new problem record is **not** added here; proposed line: `- [Lean/Lake file reads on the external APFS volume fail intermittently with EINTR, blocking the formal gate] (formal-lean-external-volume-eintr.md) - open`.
- PR creation: the keyring token for github.com is invalid on this host; a fine-grained token supplied in chat authenticates (`gh api user` -> aburan28) but `gh pr create` returns "Resource not accessible by personal access token (createPullRequest)" (no pull-request write permission). The branch `lead-discovery/20261024-mythos-ecdlp` is pushed; the PR is opened by the user with `gh pr create` after `gh auth login -h github.com`, or from the compare page, with the body in the session summary.
- Conductor: task T-5083 holds the touched paths (epoch 7); further lease renewals were refused ("retry budget exhausted (4/4)").

- Pre-existing: `tools/validate_ledger.py` reports `FAIL: 104 new validation error(s)` on origin/main records (IDEA-20261003-*, IDEA-20261016-*, experiments/EXP-AES-14352a, EXP-ICPERF-*, ...), none naming this session's records (tracked by `problems/full-ledger-validation-errors.md`); `lead_atlas.py --check --strict-paths` reports 2 errors on `LEAD-RHO-pool-class-quotient` (ecc-challenges evidence paths that exist only on branch `claude/ecdlp-leads-analysis-l7f87y`, commit 1f49962; proposed diff: add `ref: claude/ecdlp-leads-analysis-l7f87y` to those two evidence entries, or merge that branch).

i. Verdict lines

- LEAD-ICE-generic-oracle-arity-cost-floor (IDEA-20261024-a2b05d): advance | needs the repository Lean gate on a host without the EINTR fault, then a separate sheriff review, to decide the proof upgrade.
- LEAD-ICE-charged-toy-ic-exponent-vs-rho: needs the measured enumeration order k of the O5 oracle (oracle-order census spec) to decide.
- LEAD-ICP-generic-table-decomposition: advance (revisit_when sharpened to the threshold line); no disposition change.
- LEAD-ICP-extension-field-embedding (TASK-20261024-551bd9): needs the Lean gate to decide the kind upgrade; the audit's structural lemma is the stronger target.
- LEAD-ICE-sat-xor-pdp-oracle (IDEA-20261024-741c49): needs an admitted ns-per-conflict calibration on a qualified host (break-even 1/458 of a group addition) and the PROTOCOL-4 re-freeze to decide; the band is unresolved inside w in (0, 0.1), so neither branch is met today.
- LEAD-ICP-coppersmith-smallroot-semaev (IDEA-20261024-1b76a4): needs the independent review of the corner lemma and the rank-2 GAP probe to decide; predicted close (proof, conditional on H1).
- LEAD-ICP-digit-presentation-solving-degree (IDEA-20261024-e65d46, IDEA-20261024-5bbcda): needs the five-lemma proof review plus the exact PC-degree checker (direct branch, odd-order curves) and the H4w coefficient sampling at m = 6-8 (expanded presentation, rigorous at m <= 4) to decide; the chained cell and m = 5 at 128 bits stay open.
- LEAD-ICP-prime-ic-constant-engineering (IDEA-20261024-06f16f): needs the paired `mitm-full:m=3` arm against the incumbent at fp20-fp28 with matched rho, A/A and the band to decide; Tier I only.
- LEAD-ICP-isogeny-transported-relation-yield (IDEA-20261024-892d61): advance | needs the pre-registered exact occupancy tables filed in the EXP-ECTD-0cf008 amendment to decide.
- LEAD-TRF-padic-canonical-section, LEAD-TRF-small-genus-cover-general, LEAD-EXO-augmented-oracle-ggm, LEAD-EXO-miller-pairing-cocycle, LEAD-RHO-non-automorphism-equivalence, LEAD-ICE-chained-semaev-dreg-vs-semiregular: advance (row splits or sharper next steps proposed in PROPOSED-DIFFS.md 7-12, each verified by three skeptics); the Coordinator decides.
- Six proposed new rows from the fold (section b): needs their own `--match` runs and criteria blocks to decide; not filed here.

Checks

check	result
`python3 tools/validate_ledger.py`	FAIL, 104 pre-existing errors on origin/main (re-run after merging origin/main e5f46a5951 and again after filing the six scouted ideas: the same 104); 0 errors and 0 advisories name RQ-ECDLP-361f3a or any IDEA-20261024-* record of this session (every error line in `checks/validate_ledger_excerpt.txt`)
`python3 tools/lead_atlas.py --check` (plain)	PASS: `lead atlas OK: 147 leads ...; coverage 2056/2089 (33 unmapped)`; the 7 new ideas are mapped in `coverage.tsv`
`python3 tools/check_merge_hygiene.py --base origin/main`	PASS on the 27 touched paths (no conflict markers, no unparseable records, no identifier collisions)
`python3 tools/lead_atlas.py --check --strict-paths`	FAIL, 2 pre-existing errors (LEAD-RHO-pool-class-quotient evidence[4], [5]); 0 attributable to this session; without `--strict-paths` these are warnings and the check passes
`python3 tools/build_frontier_map.py --check`	PASS: `known-results map OK: 52 rows`
`lake build / AxiomAudit.lean / verify_formal_targets.py`	NOT RUN TO COMPLETION. `lake build` fails on `.trace` reads with EINTR; `verify_formal_targets.py` was run once (22:26-22:40 PDT) and stopped after 5 of about 90 targets because every receipt, including previously machine-verified Endosweep/Bin-std targets, read `formalization_blocked` / `lake build failed` with EINTR in the build log, i.e. the gate measures the host fault here. Plain `lean` compile of all three new files: PASS (logs in `checks/`). The three new targets stay undischarged.